

# Complete the Resume RAG, backend implementation and test it.

## 1.Back end → APi testing

The image displays two screenshots related to the Resume RAG project.

**Top Screenshot (VS Code):** Shows the `PHASE-IMPLEMENTATION-TRACKER.md` file. The file contains a checklist of 13 phases, all of which are marked as "COMPLETED & APPROVED" or "COMPLETED & TESTED". The phases include: Project Setup, PDF Upload API, PDF Text Extraction, Text Cleaning, Resume Parser, Regex Utilities, Skills Detection, Optional LLM Parser, Mistral Embedding Generation, MongoDB Ingestion, Resume Ingestion Service, Error Handling, and Logging. A "Complete Route Map" section shows the endpoints for Phase 2 (POST /v1/resume/upload) and Phase 3 (POST /v1/resume/extract).

**Bottom Screenshot (Postman):** Shows the "Detect Skills" endpoint being tested. The request is a POST to `localhost:3000/v1/resume/skills`. The response is a 200 OK status with a JSON body containing the following data:

```
{
  "success": true,
  "fileName": "Naukri_AbhiramIV[2y_0m].pdf",
  "skills": [
    {
      "name": "Azure",
      "category": "cloudPlatforms",
      "confidence": 100,
      "frequency": 3,
      "context": "com Address XII standard - 98% Quality Engineer Manual Testing Web Testing Selenium Java Appium GIT "
    },
    {
      "name": "Agile",
      "category": "methodologies",
      "confidence": 93,
      "frequency": 3,
      "context": "com Address XII standard - 98% Quality Engineer Manual Testing Web Testing Selenium Java Appium GIT "
    },
    {
      "name": "Devops",

```

TestingLearning

Resume Ingestion Pipeline — Complete Collection > Phase 6 — Regex Utilities > Parsing Statistics

POST `{{base_url}}/v1/resume/parsing-stats` Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

Query Params

| Key | Value | Description | Bulk Edit |
|-----|-------|-------------|-----------|
| Key | Value | Description |           |

Body Cookies Headers 9 Test Results 3/3 200 OK • 10 ms • 606 B Save Response

```
1 {
2   "success": true,
3   "fileName": "Naukri_AbixamiV[2y_0m].pdf",
4   "statistics": {
5     "emailCount": 1,
6     "phoneCount": 1,
7     "skillCount": 7,
8     "technicalSkillCount": 7,
9     "softSkillCount": 0,
10    "dateRangeCount": 5,
11    "sectionCount": 0,
12    "bulletPointCount": 0,
13    "wordCount": 376,
14    "lineCount": 1
15  },
16   "message": "Parsing statistics calculated successfully"
17 }
```

Connect Git Console Terminal Import Complete

TestingLearning

Resume Ingestion Pipeline — Complete Collection > Phase 6 — Regex Utilities > Extract Experience

POST `{{base_url}}/v1/resume/extract-experience` Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

Query Params

| Key | Value | Description | Bulk Edit |
|-----|-------|-------------|-----------|
| Key | Value | Description |           |

Body Cookies Headers 9 Test Results 2/2 200 OK • 6 ms • 1.58 KB Save Response

```
1 {
2   "success": true,
3   "fileName": "Naukri_AbixamiV[2y_0m].pdf",
4   "experience": {
5     "yearsOfExperience": [
6       "7+ Years of Experience"
7     ],
8     "jobTitles": [
9       "Engineer",
10      "Analyst"
11    ],
12    "companies": [
13      "WORK EXPERIENCESKILLSCONTACT Phone",
14      "EDUCATION",
15      "REC Engineering Colleg Bangalore",
16      "India Manual",
17      "Automation Test Engineer",
18      "Years",
19      "Experience",
20      "Web Automation",
21      "Test Strategy",
22      "Quality Assurance",
23      "Focused",
24      "Process Improvement",
25      "Delivering High",
26      "Quality Software",
27      "Email Manimekalai",
28      "Address XII",
29      "Quality Engineer Manual Testing Web Testing Selenium Java Appium GIT Azure Devops Agile JIRA Robot",
30      "Plawright R"
31    ]
32  }
33 }
```

Connect Git Console Terminal Import Complete

TestingLearning

Resume Ingestion Pipeline — Complete Collection > Phase 6 — Regex Utilities > Extract Skills (Regex)

POST `{{base_url}}/v1/resume/extract-skills` Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

Query Params

| Key | Value | Description | Bulk Edit |
|-----|-------|-------------|-----------|
| Key | Value | Description |           |

Body Cookies Headers 9 Test Results 2/2 Save Response

```
1 {
2   "success": true,
3   "fileName": "Naukri_Abizamiv[2y_0m].pdf",
4   "skills": {
5     "technicalSkills": [
6       "R",
7       "Java",
8       "Azure",
9       "Git",
10      "Jira",
11      "Jenkins",
12      "Github"
13    ],
14    "softSkills": [],
15    "allSkills": [
16      "R",
17      "Java",
18      "Azure",
19      "Git",
20      "Jira",
21      "Jenkins",
22      "Github"
23    ]
24  }
```

Connect Git Console Terminal Import Complete

Globals Vault Tools

TestingLearning

Resume Ingestion Pipeline — Complete Collection > Phase 6 — Regex Utilities > Extract Contact Info

POST `{{base_url}}/v1/resume/extract-contact` Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beautify

```
1 {
2   "text": "{{cleanedText}}",
3   "fileName": "Manimekalai CV-1780217283997-876932167.pdf"
4 }
5
```

Body Cookies Headers 9 Test Results 3/3 Save Response

```
1 {
2   "success": true,
3   "fileName": "Manimekalai CV-1780217283997-876932167.pdf",
4   "contactInfo": {
5     "emails": [
6       "ammasi@gmail"
7     ],
8     "phones": [
9       "9360226898"
10    ],
11    "linkedin": null,
12    "github": null,
13    "websites": []
14  },
15   "message": "Contact information extracted successfully"
16 }
```

Connect Git Console Terminal Import Complete

Globals Vault Tools

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Resume Ingestion Pipeline — Complete Collection > Phase 5 — Resume Parser > Parse Resume (Regex)

POST `{{base_url}}/v1/resume/parse` Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beautify

```
1 {
2   "cleanedText": "WORK EXPERIENCESKILLSCONTACT Phone 9360226898 EDUCATION 2011-2015
REC Engineering Colleg Bangalore, India Manual & Automation Test Engineer | 7+ Years of Experience in Web
Automation, Test Strategy, and Quality Assurance | Focused on Process Improvement and Delivering High-Quality
Software. Email Manimekalai. ammasi@gmail. com Address XII standard - 90% Quality Engineer Manual Testing Web
Testing Selenium Java Appium GIT Azure Devops Agile JIRA Robot framework Playwright B. E, Information Technology
70% MANIMEKALAI AMMASI Senior Test Engineer Tech Mahindra [Client-Kimberly Clark] Cognizant Technology Solutions
[Client-Viacom] Sr. Test Analyst 05-2019 to 01-2022 Benayah Solutions -[Client -Vizsafe] 2009-2011 Indian
```

Body Cookies Headers 9 Test Results 5/5 200 OK · 53 ms · 1.1 KB Save Response

`{}` JSON Preview Visualize

```
1 {
2   "success": true,
3   "fileName": "Manimekalai CV-1780217283997-876932167.pdf",
4   "parsedAt": "2026-06-19T15:04:11.797Z",
5   "confidence": 27.5,
6   "sectionsFound": [
7     "personalInfo"
8   ],
9   "personalInfo": {
10    "fullName": "",
11    "phone": "9360226898",
12    "location": "XII standard - 90% Quality Engineer Manual Testing Web Testing Selenium Java Appium GIT Azure Devops
Agile JIRA Robot framework Playwright B. E, Information Technology 70% MANIMEKALAI AMMASI Senior Test
Engineer Tech Mahindra [Client-Kimberly Clark] Cognizant Technology Solutions [Client-Viacom] Sr. Test
Analyst 05-2019 to 01-2022 Benayah Solutions -[Client -Vizsafe] 2009-2011 Indian Matriculation School
Proficient in Selenium WebDriver"
13  },
14   "experience": [],
15   "education": [],
16   "skills": [],
17   "certifications": [],
18   "languages": [],
19   "projects": [],
20   "message": "Resume parsed successfully with 27.5% confidence."
```

Connect Git Console Terminal Import Complete

Globals Vault Tools

TestingLearning

Resume Ingestion Pipeline — Complete Collection > Phase 5 — Resume Parser > Parse Resume (Regex)

POST `{{base_url}}/v1/resume/parse` Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beautify

```
1 {
2   "cleanedText": "WORK EXPERIENCESKILLSCONTACT Phone 9360226898 EDUCATION 2011-2015
REC Engineering Colleg Bangalore, India Manual & Automation Test Engineer | 7+ Years of Experience in Web
Automation, Test Strategy, and Quality Assurance | Focused on Process Improvement and Delivering High-Quality
Software. Email Manimekalai. ammasi@gmail. com Address XII standard - 90% Quality Engineer Manual Testing Web
Testing Selenium Java Appium GIT Azure Devops Agile JIRA Robot framework Playwright B. E, Information Technology
70% MANIMEKALAI AMMASI Senior Test Engineer Tech Mahindra [Client-Kimberly Clark] Cognizant Technology Solutions
[Client-Viacom] Sr. Test Analyst 05-2019 to 01-2022 Benayah Solutions -[Client -Vizsafe] 2009-2011 Indian
Matriculation School Proficient in Selenium WebDriver, TestNG, Maven, Appium, and Page Object Model (POM) for
building scalable automation frameworks. Strong expertise in test case design, execution, defect tracking, and test
planning using tools like JIRA and Azure DevOps. Skilled in functional, regression, integration, smoke, and UAT
testing, with hands-on experience in Agile/Scrum methodologies. Experience in CI/CD integration using Jenkins and
GitHub, improving test efficiency, and reducing manual effort. Conducted requirement analysis and maintained RTM to
ensure complete test coverage and traceability. Collaborated with cross-functional teams to define QA strategy,
streamline test processes, and enhance overall product quality. Passionate about improving automation processes,
optimizing QA strategy, and driving continuous quality improvement. Expertise in developing and maintaining robust
automation frameworks for web and mobile applications using Appium. Skilled in keyword-driven, data-driven, and
hybrid automation frameworks to enhance test coverage and reusability. Integrated automation scripts with Jenkins
and GitHub for continuous integration and delivery (CI/CD). Collaborated with QA and development teams to optimize
automation processes, reduce manual effort, and improve test efficiency. Hands-on experience in Manual Testing for
web and mobile applications, ensuring functional accuracy and reliability. Skilled in test case design, execution,
and defect tracking using tools like JIRA and Azure DevOps. Proficient in performing functional, regression,
integration, and UAT testing in Agile/Scrum environments. Strong in requirement analysis and preparing RTM to
ensure complete test coverage. Collaborated with cross-functional teams to deliver high-quality releases and
improve overall testing efficiency. Sr. Test Analyst 01-2022 to Present Software Test Engineer 01-2018 to 01-2019",
3   "fileName": "Manimekalai CV-1780217283997-876932167.pdf"
4 }
```

Response History

Send + Get a successful response

Send + Visualize response

Send + Write tests

Connect Git Console Terminal Import Complete

Globals Vault Tools

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COLLECTIONS

Phase 7 - Skills Detection

Phase 8 - LLM Resume Parser

Phase 9 - Mistral Embedding Generation

Postman Echo

QA Bot TS - LangChain API Copy 2

RAG MongoDB - Data Ingestion

RAG Platform - Backend (Local)

Resume AI - Phase 3 (Upload + Extract)

Resume Ingestion - Phase 2: PDF Upload API

Resume Ingestion Pipeline — Complete Collection

Phase 2 — PDF Upload

POST Upload PDF Resume

POST Upload — No File (Error 400)

Phase 3 — PDF Text Extraction

POST Extract Text from PDF

POST Extract — Missing filePath (Error 400)

Phase 4 — Text Cleaning

Phase 5 — Resume Parser

Phase 6 — Regex Utilities

Phase 7 — Skills Detection

Phase 8 — LLM Parser (Optional)

Phase 9 — Mistral Embedding

Phase 10 — MongoDB Ingestion

Phase 11 — Full Ingestion Pipeline

Phase 12 — Error Handling

Phase 13 — Logging

Resumes\_AI-Rag

xsbxhs

ENVIRONMENTS

SPECS

FLOWs

POST Upd

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Resume Ingestion Pipeline — Complete Collection > Phase 3 — PDF Text Extraction > Extract Text from PDF

Save

Share

POST

(((base\_url)) /v1/resume/extract)

Send

Docs

Params

Authorization

Headers 10

Body

Scripts

Settings

Cookies

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

Schema

Beautify

1

2

3

4

{

"filePath": "/Users/meghavela/Desktop/Genai Assignments/week9/resume-ai-rag/uploads/Manimekalai

CV-1780217283997-876932167.pdf",

"fileName": "Manimekalai CV-1780217283997-876932167.pdf"

}

Body

Cookies

Headers 9

Test Results 4/4

200 OK

173 ms

3.41 KB

Save Response

⋮

{}

JSON

Preview

Visualize

execution, defect tracking, and test planning using tools like JIRA and Azure DevOps. Skilled in functional, regression, integration, smoke, and UAT testing, with hands-on experience in Agile/Scrum methodologies. Experience in CI/CD integration using Jenkins and GitHub, improving test efficiency, and reducing manual effort. Conducted requirement analysis and maintained RTM to ensure complete test coverage and traceability. Collaborated with cross-functional teams to define QA strategy, streamline test processes, and enhance overall product quality. Passionate about improving automation processes, optimizing QA strategy, and driving continuous quality improvement. Expertise in developing and maintaining robust automation frameworks for web and mobile applications using Appium. Skilled in keyword-driven, data-driven, and hybrid automation frameworks to enhance test coverage and reusability. Integrated automation scripts with Jenkins and GitHub for continuous integration and delivery (CI/CD). Collaborated with QA and development teams to optimize automation processes, reduce manual effort, and improve test efficiency. Hands-on experience in Manual Testing for web and mobile applications, ensuring functional accuracy and reliability. Skilled in test case design, execution, and defect tracking using tools like JIRA and Azure DevOps. Proficient in performing functional, regression, integration, and UAT testing in Agile/Scrum environments. Strong in requirement analysis and preparing RTM to ensure complete test coverage. Collaborated with cross-functional teams to deliver high-quality releases and improve overall testing efficiency. Sr.Test Analyst 01-2022 to Present Software Test Engineer 01-2018 to 01-2019\n",

14

15

16

17

18

19

"metadata": {

"info": {},

"extractedAt": "2026-06-19T15:01:34.550Z"

},

"message": "Text extracted successfully from PDF."

}

Connect Git

Console

Terminal

Import Complete

Globals

Vault

Tools

⚙️

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TestingLearning

Resume Ingestion Pipeline — Complete Collection > Phase 2 — PDF Upload > Upload PDF Resume

POST `POST` `[[base_url]]/v1/resume/upload?` Send

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

☐ none ☒ form-data ☐ x-www-form-urlencoded ☐ raw ☐ binary ☐ GraphQL

| Key    | Value                                       | Description              | Bulk Edit   |
|--------|---------------------------------------------|--------------------------|-------------|
| resume | File <code>Naukri_AbirmiV[2y_0m].pdf</code> | Select a PDF resume file |             |
| Key    | Text                                        | Value                    | Description |

Body Cookies Headers 9 Test Results 3/3 200 OK - 13 ms - 711 B Save Response

```
1 {
2   "success": true,
3   "fileName": "Naukri_AbirmiV[2y_0m].pdf",
4   "originalName": "Naukri_AbirmiV[2y_0m].pdf",
5   "filePath": "/Users/meghavela/Desktop/Genai Assignments/week9/resume-ai-rag/uploads/Naukri_AbirmiV[2y_0m].pdf",
6   "fileSize": 62071,
7   "mimeType": "application/pdf",
8   "uploadedAt": "2026-06-19T14:58:43.628Z",
9   "message": "File uploaded successfully. Ready for processing."
10 }
```

TestingLearning

Resume Ingestion Pipeline — Complete Collection > Phase 3 — PDF Text Extraction > Extract Text from PDF

POST `POST` `[[base_url]]/v1/resume/extract` Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beautify

```
1 {
2   "filePath": "/Users/meghavela/Desktop/Genai Assignments/week9/resume-ai-rag/uploads/Manimekalai CV-1786217283997-876932167.pdf",
3   "fileName": "Manimekalai CV-1786217283997-876932167.pdf"
4 }
```

Body Cookies Headers 9 Test Results 4/4 200 OK - 173 ms - 3.41 KB Save Response

```
1 {
2   "filePath": "/Users/meghavela/Desktop/Genai Assignments/week9/resume-ai-rag/uploads/Manimekalai CV-1786217283997-876932167.pdf",
3   "fileName": "Manimekalai CV-1786217283997-876932167.pdf"
4 }
```

14 "metadata": {  
15 "info": {},  
16 "extractedAt": "2026-06-19T15:01:34.550Z"  
17 },  
18 "message": "Text extracted successfully from PDF."  
19 }

TestingLearning

Resume Ingestion Pipeline — Complete Collection > Phase 8 — LLM Parser (Optional) > LLM Parse Resume

POST `{{base_url}} /v1/resume/llm-parse` Send

Docs Params Authorization Headers 10 Body ● Scripts ● Settings Cookies

Body Cookies Headers 9 Test Results 4/4 200 OK 2.67 s 3.01 KB Save Response

```
{
  "startDate": "05-2019",
  "endDate": "01-2022",
  "currentlyWorking": false,
  "responsibilities": [
    "Proficient in Selenium WebDriver, TestNG, Maven, Appium, and Page Object Model (POM) for building scalable automation frameworks.",
    "Strong expertise in test case design, execution, defect tracking, and test planning using tools like JIRA and Azure DevOps.",
    "Skilled in functional, regression, integration, smoke, and UAT testing, with hands-on experience in Agile/Scrum methodologies.",
    "Experience in CI/CD integration using Jenkins and GitHub, improving test efficiency, and reducing manual effort.",
    "Conducted requirement analysis and maintained RTM to ensure complete test coverage and traceability.",
    "Collaborated with cross-functional teams to define QA strategy, streamline test processes, and enhance overall product quality."
  ]
},
{
  "jobTitle": "Sr. Test Analyst",
  "company": "Cognizant Technology Solutions [Client-Viacom]",
  "startDate": "",
  "endDate": "",
  "currentlyWorking": false,
  "responsibilities": []
},
{
  "jobTitle": "Sr. Test Analyst",
  "company": "Benayah Solutions [Client-Vizsafe]",
  "startDate": "01-2022",
  "endDate": "Present",
  "currentlyWorking": true,
  "responsibilities": []
},
{
  "jobTitle": "Software Test Engineer",
  "company": ""
}
```

TestingLearning

Resume Ingestion Pipeline — Complete Collection > Phase 8 — LLM Parser (Optional) > LLM Parse Resume

POST `{{base_url}} /v1/resume/llm-parse` Send

Docs Params Authorization Headers 10 Body ● Scripts ● Settings Cookies

Body Cookies Headers 9 Test Results 4/4 200 OK 2.67 s 3.01 KB Save Response

```
{
  "success": true,
  "enabled": true,
  "fileName": "Naukri_AbiramV[2y_0m].pdf",
  "parsedData": {
    "name": "Manimekalai Ammasi",
    "email": "manimekalai.ammasi@gmail.com",
    "phone": "9360226898",
    "location": "",
    "linkedin": "",
    "github": "",
    "summary": "Manual & Automation Test Engineer | 7+ Years of Experience in Web Automation, Test Strategy, and Quality Assurance | Focused on Process Improvement and Delivering High-Quality Software.",
    "skills": [
      "Manual Testing",
      "Web Testing",
      "Selenium",
      "Java",
      "Appium",
      "GIT",
      "Azure Devops",
      "Agile",
      "JIRA",
      "Robot Framework",
      "Playwright",
      "Quality Assurance",
      "Test Strategy",
      "Web Automation",
      "Process Improvement",
      "Selenium WebDriver",
      "TestNG",
      "Maven",
      "Page Object Model (POM)",
      "Jenkins",
      "GitHub"
    ]
  }
}
```

[illegible]